Customer Purchasing Behavior and Product Recommendation Report

1. Project Overview

This report analyzes Amazon customer behavior to enhance personalized shopping experiences and optimize marketing strategies. The primary objective is to build a data-driven understanding of how demographics and browsing habits influence purchase frequency and satisfaction.

2. Data Cleaning and Preparation (Task 1)

To ensure high-quality insights, the following steps were taken:

- **Duplicate Removal**: Identical survey entries were removed to prevent bias.
- **Standardization**: Categorical fields like 'Gender' and 'Purchase Frequency' were standardized for consistency.
- **Missing Values**: Handled nulls in fields like `Product_Search_Method` to maintain dataset integrity.
- **Data Conversion**: Textual ratings for `Shopping_Satisfaction` and `Customer_Reviews_Importance` were converted into numerical formats (1-5 scale) for statistical analysis.

Task 1

```python
# 1.1 Remove duplicate or inconsistent survey responses
df = df.drop_duplicates()
```

```python
# 1.2 Standardize categorical entries
# Example: Standardizing Gender and Frequency
df['Gender'] = df['Gender'].str.strip().str.capitalize()
df['Purchase_Frequency'] = df['Purchase_Frequency'].str.lower()
```

```python
# 1.3 Handle missing values in Product_Search_Method and others
df['Product_Search_Method'] = df['Product_Search_Method'].fillna('Unknown')
# Dropping rows where essential target data is missing
df.dropna(subset=['Shopping_Satisfaction'], inplace=True)
```

```python
# 1.4 Rename duplicate or misformatted columns
# Removing trailing spaces in Rating_Accuracy and handling the duplicate Recommendation column
df.rename(columns={'Rating_Accuracy ': 'Rating_Accuracy'}, inplace=True)
# If a duplicate column exists, we drop the second one
df = df.loc[:, ~df.columns.duplicated()]
```

```python
# 1.5 Convert numerical rating columns to appropriate numeric types
cols_to_fix = ['Customer_Reviews_Importance', 'Shopping_Satisfaction', 'Rating_Accuracy']
for col in cols_to_fix:
    df[col] = pd.to_numeric(df[col], errors='coerce')
```

3. Descriptive Behavior Analysis (Task 2)

Key metrics derived from the survey data:

- **Demographics**: The customer base was segmented by age and gender distribution to identify primary user groups.
- **Purchase Trends**: Most popular product categories (e.g., Clothing, Electronics, Beauty) were identified alongside the average purchase frequency.
- **Friction Points**: Common cart abandonment factors, such as price and shipping costs, were pinpointed.

Task 2

```python
import matplotlib.pyplot as plt
import seaborn as sns
```

```python
# 2.1 Summarize customer demographics (age, gender distribution)
print(df['age'].describe())
print(df['Gender'].value_counts())
```

```python
# 2.2 Analyze purchase frequency and popular categories
print(df['Purchase_Frequency'].value_counts())
# Assuming categories are comma-separated in the survey
top_categories = df['Purchase_Categories'].str.get_dummies(sep=',').sum().sort_values(ascending=False)
print(top_categories.head(5))
```

```python
# 2.3 Top browsing methods and cart abandonment factors
print(df['Product_Search_Method'].value_counts())
print(df['Cart_Abandonment_Factors'].value_counts())
```

```python
# 2.4 Calculate mean/median for satisfaction and accuracy
metrics = df[['Shopping_Satisfaction', 'Rating_Accuracy']].agg(['mean', 'median'])
print(metrics)
```

```python
# 2.5 Generate summary statistics and visualizations
df.hist(bins=15, figsize=(15, 10))
plt.show()
```

4. Customer Segmentation (Task 3)

Using **K-Means Clustering**, customers were grouped into three distinct profiles:

- **Frequent Buyers**: Users with high purchase frequency and high satisfaction scores.
- **Occasional Shoppers**: Users with medium frequency and moderate satisfaction.
- **At-Risk Customers**: Users with low satisfaction or high cart abandonment rates who require targeted promotions.

Task 3: Customer Segmentation and Profiling

```
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
```

```
# 3.1 & 3.5 Use clustering (K-Means) for behavioral grouping
# Selecting features for segmentation (Frequency and Satisfaction)
# Note: Frequency needs to be encoded numerically first
freq_map = {'few times a week': 4, 'once a month': 2, 'less than once a month': 1}
df['Freq_Score'] = df['Purchase_Frequency'].map(freq_map).fillna(0)

features = df[['Freq_Score', 'Shopping_Satisfaction']]
scaler = StandardScaler()
scaled_features = scaler.fit_transform(features)

kmeans = KMeans(n_clusters=3, random_state=42)
df['Segment'] = kmeans.fit_predict(scaled_features)
```

```
# 3.2 - 3.4 Define profiles based on cluster centers
# Segment 0: Frequent Buyers, Segment 1: Occasional, Segment 2: At-Risk
```

5. Recommendation and Review Insights (Task 4)

Analysis of Amazon’s recommendation and review systems revealed:

- **Trust Impact**: A positive correlation exists between "Review Reliability" and "Rating Accuracy," suggesting that customers who trust reviews are more satisfied with their purchases.
- **Recommendation Helpfulness**: Users who find personalized recommendations helpful generally report higher overall shopping satisfaction.
- **Actionable Insight**: Enhancing the transparency of product ratings can significantly boost customer loyalty.

Task 4

```
# 4.1 Relationship between recommendation helpfulness and satisfaction [cite: 37]
sns.boxplot(x='Recommendation_Helpfulness', y='Shopping_Satisfaction', data=df)
plt.title('Helpfulness vs Satisfaction')
plt.show()
```

```
# Mapping text responses to numerical values for correlation analysis
reliability_map = {
    'Heavily': 3,
    'Moderately': 2,
    'Occasionally': 1,
    'Rarely': 0,
    'Never': 0
}

# Create a new numerical column based on the mapping
df['Review_Reliability_Num'] = df['Review_Reliability'].map(reliability_map).fillna(0)

# Now calculate the correlation with Rating_Accuracy (which is already numeric from Task 1)
correlation_value = df['Review_Reliability_Num'].corr(df['Rating_Accuracy'])

print(f"Correlation between Review Reliability and Rating Accuracy: {correlation_value:.2f}")
```

```
# 4.3 Trends in personalized recommendation trust
trust_trend = df['Personalized_Recommendation_Frequency'].value_counts()
print(trust_trend)
```

6. Visual Dashboard Summary (Task 5)

The following trends were visualized for the leadership team:

- **Browsing Frequency**: A pie chart illustrating how often users browse without purchasing.
- **Satisfaction Distribution**: A bar chart highlighting that most users fall within the 3-5 rating range.
- **Correlation Heatmap**: Visual proof of the relationship between system features and user retention.

Task 5: Visualization and Reporting

```
# 5.1 & 5.2 Purchase categories bar chart
top_categories.plot(kind='bar', color='skyblue')
plt.title('Top Purchase Categories')
plt.show()
```

```
# 5.3 Browsing frequency distribution
df['Browsing_Frequency'].value_counts().plot(kind='pie', autopct='%1.1f%%')
plt.title('Browsing Frequency')
plt.show()
```

```
# Task 5.5: Visualization of Satisfaction Levels
plt.figure(figsize=(8, 5))
sns.countplot(x='Shopping_Satisfaction', data=df, palette='viridis')
plt.title('Distribution of Customer Shopping Satisfaction Levels')
plt.xlabel('Satisfaction Rating (1-5)')
plt.ylabel('Number of Customers')
plt.show()
```

```
# Task 5.6: Correlation Heatmap
# We select only the numeric columns to avoid the "could not convert string to float" error.
# We include the newly created 'Recommendation_Helpfulness_Num' and 'Shopping_Satisfaction'
numeric_cols = df.select_dtypes(include=[np.number])

plt.figure(figsize=(10, 8))
# Use numeric_only=True to ensure the function only looks at numbers
sns.heatmap(numeric_cols.corr(), annot=True, cmap='coolwarm', fmt=".2f")
plt.title('Correlation Heatmap: Recommendations vs. Satisfaction')
plt.show()
```

7. Final Recommendations

1. **Personalization**: Increase the frequency of personalized recommendations for "Occasional Shoppers" to move them into the "Frequent Buyer" segment.

2. **Abandonment Recovery**: Implement targeted discounts or notifications for "At-Risk Customers" based on their specific cart abandonment factors.

3. **Review Verification**: Improve the "Review Helpfulness" system to ensure high-quality feedback is prioritized, as this drives overall satisfaction.

Video Link :
https://drive.google.com/drive/folders/1EH3btex1rYZ5Eeg_UZ6JphBj5qwp_DDr?usp=sharing